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NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY**THIRD SEMESTER-B.TECH.(ECE/EIOT)****MID-SEMESTER EXAMINATION, September, 2023****Course Code: ECECC06/EIECC06 Course Title: Probability Theory and Random Processes****Time: 1.5 Hours****Max. Marks: 25****Note: Attempt all questions. Missing data/information(if any), may be suitably assumed & mentioned in the answer.**

S.No.	Question	Marks	CO
1.	Attempt all parts		
(a)	Calculate mean, mean square value and variance of uniform distribution over $(0, \pi)$.	2	CO1
(b)	Binary data are transmitted over a noisy communication channel in block of 16 binary digits. The probability that a received digit is in error due to channel noise is 0.01. assume that the errors occurring in various digit positions within a block are independent. (i) Find the mean errors per block. (ii) Find the variance of the number of errors per block. (iii) Find the probability that the number of errors per block is greater than or equal to 4.	3	CO1
2.	Attempt all parts		
(a)	Let X and Y be Joint continuous random variables with joint PDF is given below $f_{XY}(x, y) = \begin{cases} 2 & x, y > 0 \text{ and } x + y \leq 1 \\ 0 & \text{otherwise} \end{cases}$ Find marginal PDF and co-variance of X and Y.	3	CO2
(b)	Discuss the joint-probability distribution function (PDF) and joint-commutative distribution function (CDF), Also discuss the properties of CDF and PDF.	2	CO2
3	Attempt all parts		
(a)	Suppose that earthquakes occur with $\lambda = 2$ and with 1 week as the unit of time. (i) Find the probability that at least 3 earthquakes occur during the next 2 weeks. (ii) Find the probability distribution of the time, starting from now, until the next earthquake.	2	CO1
(b)	Let X have distribution	3	CO1